Heinrich Weber
Lehrbuch der Algebra
Working English Translation Batch 04
Abelian Groups: Bases and Invariants

Translator's note

This working English translation is prepared from the cleaned Kimi-typeset Weber source currently in hand. It continues the group-theory opening of the second volume and covers the beginning of the second section on Abelian groups, namely the material on representation by a basis and on the invariants of an Abelian group. I have preserved the mathematical order of Weber's argument while lightly modernizing punctuation and English phrasing.

Second Section. Abelian groups

§9. Representation of Abelian groups by a basis

We have already called those groups in which the commutative law holds for composition *commutative* or *Abelian* groups. For these groups, which are of particular importance in applications, the laws are much simpler than in general groups, as we now see.

Let S be an Abelian group of order n. We shall show that the elements of S can be represented in the form

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \tag{1}$$

where $A_1, \ldots, A_\nu$ are suitable elements of the group and the exponents $\alpha_1, \ldots, \alpha_\nu$ run through complete residue systems modulo positive integers $a_1, \ldots, a_\nu$, the orders of $A_1, \ldots, A_\nu$. Such a system $A_1, \ldots, A_\nu$ is called a *basis* of the Abelian group.

One sees immediately that every element of the form (1) actually occurs: for example, one obtains A_i by taking $\alpha_i = 1$ and $\alpha_k = 0$ for $k \neq i$.

It remains to prove that every element occurs equally often. Suppose

$$1 = A_1^{h_1} A_2^{h_2} \cdots A_n^{h_n} \tag{2}$$

is a representation of the identity. Then the value of θ in (1) is unchanged if one replaces $\alpha_1, \ldots, \alpha_n$ by $\alpha_1 + h_1, \ldots, \alpha_n + h_n$. Hence each element θ is represented at least as many times by (1) as the identity is represented by (2).

Conversely, if two exponent systems $\alpha_1, \ldots, \alpha_n$ and $\alpha'_1, \ldots, \alpha'_n$ yield the same element θ, then

$$1 = A_1^{\alpha_1 - \alpha'_1} A_2^{\alpha_2 - \alpha'_2} \cdots A_n^{\alpha_n - \alpha'_n}. \tag{3}$$

Thus the difference of the two exponent systems gives a representation of the identity. Therefore the number of representations of θ can be no larger than the number of representations of 1.

If we denote that number by h, then the total number of possible exponent choices in (1) is $a_1 a_2 \cdots a_n$, while n is the number of distinct group elements. Hence

$$nh = a_1 a_2 \cdots a_n. \tag{4}$$

From formula (4) one obtains:

Proposition 2. If r is a prime dividing the order n of the group S, then S contains an element of order r.

Indeed, one of the orders $a_1, \ldots, a_n$ must be divisible by r. If, say, $a_1 = rk$, then A_1^k has order r.

Proposition 3. If $A, B, \ldots$ are arbitrary elements of S with orders $a, b, \ldots$, then the product $AB \cdots$ again lies in S, and its order divides the least common multiple m of $a, b, \ldots$.

For

$$(AB \cdots)^m = A^m B^m \cdots = 1,$$

so m is a multiple of the order of $AB \cdots$.

Proposition 4. Suppose the order n of the group is factored as

$$n = ab$$

with a and b relatively prime. Then there exist exactly a elements A whose order divides a, and exactly b elements B whose order divides b, and every element of the group is represented uniquely in the form

$$\theta = AB. \tag{5}$$

To see this, let $\mathcal{A}$ be the set of elements whose order divides a. By Proposition 3 this is a subgroup of S. Similarly the set $\mathcal{B}$ of elements whose order divides b is a subgroup.

Let a' be the order of $\mathcal{A}$, and b' the order of $\mathcal{B}$. Then a' is relatively prime to b: if a prime r divided a', then by Proposition 2 the subgroup $\mathcal{A}$ would contain an element of order r, and since every element of $\mathcal{A}$ has order dividing a, that prime r would divide a, not b. Similarly b' is relatively prime to a.

Now choose integers x, y such that

$$ax + by = 1. \tag{6}$$

For any element $\theta \in S$ we have

$$\theta = \theta^{ax} \theta^{by}. \tag{7}$$

Since $(\theta^{ax})^b = 1$, the element θ^{ax} lies in $\mathcal{A}$, and similarly $\theta^{by} \in \mathcal{B}$. Therefore every element has a representation

$$\theta = AB. \tag{8}$$

This representation is unique. If $AB = A'B'$, then by raising to the suitable powers $by = 1 - ax$ and $ax = 1 - by$ one obtains $A = A'$ and hence $B = B'$. The number of products AB is therefore $a'b'$, and since every element of S occurs once, we have

$$n = ab = a'b'.$$

Because a is relatively prime to b' and b is relatively prime to a', it follows that

$$a = a', \qquad b = b'.$$

This proves Proposition 4 in full.

If the two subgroups $\mathcal{A}, \mathcal{B}$ are each represented by bases, then formula (8) shows that S itself is represented by a basis, namely by adjoining the basis elements of $\mathcal{A}$ and $\mathcal{B}$.

Repeating this decomposition whenever a and b can themselves be split into relatively prime factors, we reduce matters to the case where the order of the group is a power of a single prime:

$$n = p^{\mu}. \tag{9}$$

Thus it remains to prove that every Abelian group of prime-power order can be represented by a basis. Let A be an element of maximal order a. Then a is a power of p, and the order of every element of S divides a. Hence for every $\theta \in S$,

$$\theta^{a} = 1. \tag{10}$$

The elements

$$1, A, A^2, \ldots, A^{a-1} \tag{11}$$

are all distinct and form a subgroup of S. If they already exhaust the group, then S is represented by the one-element basis A.

Otherwise, for every $\theta \in S$ there exists a least positive integer b such that

$$\theta^{b} = A^{t} \tag{12}$$

for some integer t. This b divides a, hence is a power of p, and it also divides the order of θ. From (12) one finds that t must be divisible by b: writing $a = qb + b'$ with $0 \leq b' < b$, one checks that if $b' \neq 0$ then $\theta^{b'}$ would already be a power of A, contradicting the minimality of b.

Writing $a/b = k$ and setting

$$B = \theta A^{-tk}, \tag{13}$$

we obtain

$$B^{b} = 1, \tag{14}$$

and b is the least exponent with this property. Thus the elements

$$A^{\alpha_1} B^{\alpha_2}, \qquad \alpha_1 = 0, 1, \ldots, a-1, \quad \alpha_2 = 0, 1, \ldots, b-1, \tag{15}$$

are all distinct and form a subgroup S' represented by the basis A, B. If $S' = S$, we are done; otherwise the construction continues inductively.

Proceeding by induction, one obtains a subgroup $S_{\nu-1}$ with basis

$$S_{\nu-1} = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_{\nu-1}^{\alpha_{\nu-1}}, \tag{16}$$

such that:

1. the orders satisfy
$$a_1 \geq a_2 \geq \cdots \geq a_{\nu-1},$$
2. for every element $\theta \in S$, the power $\theta^{a_{\nu-1}}$ lies in $S_{\nu-1}$, and the corresponding exponents are all divisible by $a_{\nu-1}$.

If $S_{\nu-1} \neq S$, choose $\theta \in S$ so that the least positive exponent a_ν with $\theta^{a_\nu} \in S_{\nu-1}$ is as large as possible. Then $a_\nu \leq a_{\nu-1}$, and a_ν is again a power of p. Writing

$$\theta^{a_\nu} = A_1^{h_1 a_\nu} A_2^{h_2 a_\nu} \cdots A_{\nu-1}^{h_{\nu-1} a_\nu}, \tag{17}$$

define

$$A_\nu = \theta A_1^{-h_1} A_2^{-h_2} \cdots A_{\nu-1}^{-h_{\nu-1}}. \tag{18}$$

Then

$$A_\nu^{a_\nu} = 1, \tag{19}$$

and a_ν is the smallest such exponent. Consequently, the elements

$$A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \qquad 0 \le \alpha_i < a_i, \tag{20}$$

are all distinct and form a larger subgroup S_ν, again satisfying the inductive conditions.

Continuing this process, one eventually obtains the whole group S represented by a basis. This proves the theorem that every finite Abelian group can be represented by a basis.

It follows at once that every Abelian group is metacyclic. Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu},$$

and if p is a prime dividing the order a_1 of A_1, then the elements

$$\theta' = A_1^{p\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}$$

form a subgroup of index p, represented by the basis $A_1^p, A_2, \ldots, A_\nu$. Repeating this procedure shows that S admits a chain of subgroups of prime index, hence is metacyclic.

§10. The invariants of Abelian groups

The proof just given for the existence of a basis also provides a method for finding one in which the orders of the basis elements are prime powers. Nevertheless, the same Abelian group can admit different bases.

For example, let A and B have relatively prime orders a and b. Then they determine a two-element basis with group

$$A^\alpha B^\beta, \qquad \alpha = 0, 1, \ldots, a-1, \quad \beta = 0, 1, \ldots, b-1. \tag{1}$$

But the same group can also be represented by the one-element basis AB, namely in the form

$$(AB)^s, \qquad s = 0, 1, \ldots, ab-1. \tag{2}$$

Indeed, given arbitrary α, β, one can determine s modulo ab so that

$$s \equiv \alpha \pmod{a}, \qquad s \equiv \beta \pmod{b},$$

and then (2) reproduces (1).

Even so, there is a precise sense in which the structure of the basis is completely determined by the group itself.

Theorem 0.2. Suppose

$$A_1, A_2, \ldots, A_\nu$$

with orders

$$a_1, a_2, \ldots, a_\nu$$

and

$$B_1, B_2, \ldots, B_\mu$$

with orders

$$b_1, b_2, \ldots, b_\mu$$

are two bases of the same Abelian group S of order n. Let p be a prime dividing n, and write

$$a_i = p^{\alpha_i} \bar{a}_i, \qquad b_j = p^{\beta_j} \bar{b}_j,$$

where $\bar{a}_i, \bar{b}_j$ are not divisible by p. Then the prime powers p^{α_i} greater than 1 occur among the p^{β_j}, and conversely.

To prove this, order the basis elements so that

$$a_1 \leq a_2 \leq \cdots \leq a_\nu, \qquad b_1 \leq b_2 \leq \cdots \leq b_\mu. \tag{3}$$

Let m be the least common multiple of $a_1, \ldots, a_\nu$. Since the $\bar{a}_i$ are not divisible by p, the number m is also not divisible by p.

For an arbitrary element

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \tag{4}$$

one then has

$$\theta^{p^{\alpha_\nu} m} = 1.$$

Applying this to the basis elements $B_1, \ldots, B_\mu$, one sees that every b_j divides $p^{\alpha_\nu} m$. Hence p^{β_μ} divides p^{α_ν}, and the least common multiple of the b_j is again m. Reversing the rôles of the two bases, one obtains the opposite divisibility, and therefore

$$p^{\alpha_\nu} = p^{\beta_\mu}. \tag{5}$$

Assume inductively that

$$p^{\alpha_1} = p^{\beta_1}, \quad p^{\alpha_2} = p^{\beta_2}, \quad \ldots, \quad p^{\alpha_{s-1}} = p^{\beta_{s-1}} \tag{6}$$

have already been proved. Then for any element θ,

$$\theta^{p^{\alpha_s} m} = A_1^{\alpha_1 p^{\alpha_s} m} A_2^{\alpha_2 p^{\alpha_s} m} \cdots A_\nu^{\alpha_\nu p^{\alpha_s} m}. \tag{7}$$

Counting the number of distinct elements of this form yields

$$q = \frac{p^{\alpha_1}}{p^{\alpha_s}} \cdot \frac{p^{\alpha_2}}{p^{\alpha_s}} \cdots \frac{p^{\alpha_{s-1}}}{p^{\alpha_s}}. \tag{9}$$

If one now expresses the same elements in the B-basis and assumes

$$p^{\alpha_s} > p^{\beta_s}, \tag{10}$$

then the counting argument forces p^{β_s} to divide p^{α_s}, which contradicts (10) unless

$$p^{\alpha_s} = p^{\beta_s}. \tag{14}$$

This proves Theorem 0.2.

The prime powers occurring in the orders of a basis are therefore independent of the choice of basis. We call them the *invariants* of the Abelian group. Their product is equal to the order n of the group.

By §9 we can choose a basis whose element orders are prime powers. Hence those orders are exactly the invariants and are completely determined by the group.

That the invariants completely determine the group as well follows from the next theorem.

Theorem 0.3. Two Abelian groups with the same invariants are isomorphic, and isomorphic Abelian groups have the same invariants.

Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}$$

are the elements of one group S, and

$$\theta' = B_1^{\alpha_1} B_2^{\alpha_2} \cdots B_\nu^{\alpha_\nu}$$

those of another group S', with the same exponent ranges modulo the same basis orders $a_1, \ldots, a_\nu$, then we obtain an isomorphism by letting θ correspond to θ' whenever the exponent systems are identical.

Conversely, if two groups are isomorphic, then the images of a basis of one group form a basis of the other, and therefore the invariants must agree in both.